TPHY303L

ANALOG AND DIGITAL ELECTRONICS

PROJECT REPORT

GAS LEAK DETECTION AND SMART WEIGHT MONITORING FOR LPG CYLINDERS

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DECLARATION

I hereby declare that the report entitled "GAS LEAK DETECTION AND SMART WEIGHT MONITORING FOR LPG CYLINDERS" submitted by us, for the course TPHY303L - Analog and Digital Electronics to Vellore Institute of Technology is a record of bonafide work carried out by us under the supervision of Prof. Leema Rose Viannie. We further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for any other courses in the Institute or any other Institute or University

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Place: Vellore

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We would like to express my deepest gratitude to Professor Leema Rose Viannie for her invaluable guidance and support throughout our project. Her expert advice, critical feedback, and dedication have been instrumental in helping our achieve my goals.

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ABSTRACT

There is a large number of LPG accidents in India, that need to be addressed at a systematic and technical level. The project temporarily ignores the systematic attempts to tackle the issue at a technical level. The primary objective of the device is to detect the gas leak and to do so at the lowest possible price. This ensures the capability of sensing LPG leakages reaches the masses via its affordability.

INTRODUCTION

Liquefied Petroleum Gas (LPG) is a commonly used fuel due to its high energy efficiency and ease of use in domestic and industrial applications. However, despite these advantages, it presents significant safety risks, as even minor leaks can lead to hazardous situations such as fires or explosions. Since such leaks are not always immediately noticeable, there is a growing need for systems that can detect and respond to them promptly.

To address this concern, a monitoring system is designed to identify the presence of leaked gas and provide immediate user awareness. The system continuously analyzes the surrounding environment using suitable sensors to detect any unusual increase in gas concentration. Once the detected level exceeds a set limit, a notification is transmitted through a Telegram-based alert mechanism, allowing users to take quick action regardless of their location. In addition to leak detection, the system also includes a method for observing cylinder usage through weight measurement. By tracking changes in weight, it provides an estimate of the remaining gas and indicates when the cylinder is nearing empty. This helps users prepare for replacement in advance and avoid sudden interruptions. The combination of continuous monitoring and timely alerts makes the system a reliable solution to improve safety and ensure efficient LPG usage.

LITERATURE REVIEW

Patil, S. et al. [1] proposed a system for detecting LPG gas leakage using gas sensors and triggering an alarm to alert users in case of abnormal gas concentration.

Kumar, R. et al. [2] developed an IoT-based gas leakage detection system that sends real-time alerts to users through mobile applications, improving safety and remote monitoring.

Singh, A. et al. [3] designed a smart gas monitoring system using wireless communication to detect gas leakage and notify users through automated messaging systems.

Reddy, P. et al. [4] implemented a gas leakage detection system integrated with GSM technology to send SMS alerts when gas levels exceed a predefined threshold.

Sharma, V. et al. [5] proposed a system using gas sensors and microcontrollers to continuously monitor gas concentration and activate safety measures in hazardous conditions.

Gupta, N. et al. [6] developed a smart LPG monitoring system that combines gas detection with IoT platforms to provide real-time data and remote alerts.

Verma, S. et al. [7] designed a load cell-based monitoring system to measure the weight of LPG cylinders and estimate the remaining gas level for timely replacement.

Khan, M. et al. [8] proposed a system integrating gas sensors and cloud-based monitoring to track gas leakage and improve response time in emergency situations.

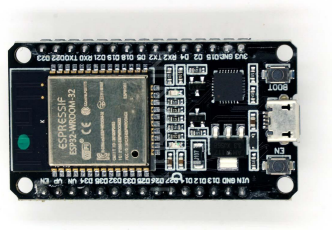
Das, R. et al. [9] developed a real-time gas monitoring system using embedded systems to detect leakage and provide early warning alerts to users.

Mehta, P. et al. [10] designed an integrated LPG monitoring system combining leakage detection and cylinder weight measurement to enhance safety and efficiency in gas usage.

Rajya Sabha Unstarred Question No.763 Answered on 02/12/2024 Data on accidents caused by LPG Gas Cylinder

ASSETS REQUIRED FOR THE PROJECT

ESP32 WROOM Development Module:

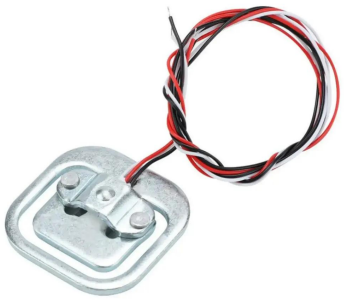


The ESP32 WROOM development module is a powerful and cost-effective microcontroller equipped with integrated Wi-Fi and Bluetooth capabilities. It is widely used in Internet of Things (IoT) applications due to its high processing performance and reliable connectivity. In this project, the ESP32 serves as the central control unit, enabling real-time data processing and communication between sensors and the user interface.

MQ-6 Gas Sensor:

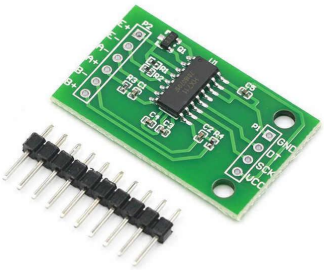
The MQ-6 gas sensor is specifically designed for the detection of LPG and other combustible gases in the surrounding environment. It operates based on variations in resistance when exposed to gas concentrations, producing an output signal that can be analyzed by the microcontroller. Owing to its high sensitivity to LPG, the MQ-6 sensor is well-suited for gas leak detection and safety applications.

50 kg Load Sensor:



The 50 kg load sensor is used to measure the weight of the LPG cylinder. It functions on the principle of strain gauges, where mechanical deformation caused by applied weight results in a proportional change in electrical resistance. This allows accurate measurement of the cylinder's weight, which is utilized to estimate the remaining gas level.

HX711 24-bit Signal Amplifier:

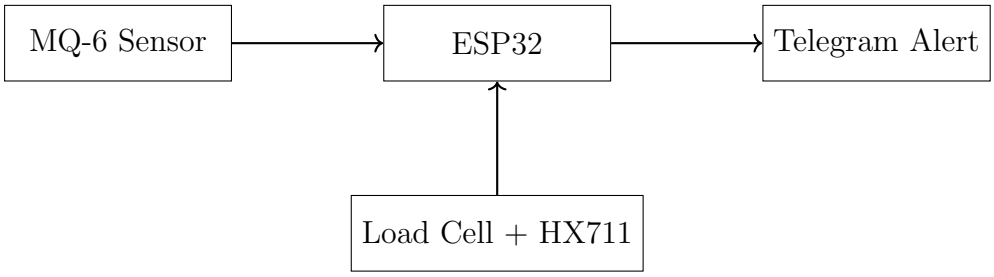


The HX711 is a high-precision 24-bit analog-to-digital converter designed for load cell applications. It amplifies the low-level electrical signals generated by the load sensor and converts them into digital data that can be processed by the microcontroller. This ensures high accuracy and stability in weight measurement.

Software:

The Arduino Integrated Development Environment (IDE) is used for writing, compiling, and uploading code to the ESP32 module. It provides a convenient platform for developing embedded applications and supports a wide range of libraries for sensor interfacing and communication. A Telegram bot is utilized as a communication interface to deliver real-time notifications to the user. When gas leakage is detected or the cylinder weight falls below a predefined threshold, alerts are sent instantly, enabling remote monitoring and timely response.

BLOCK DIAGRAM



METHODOLOGY

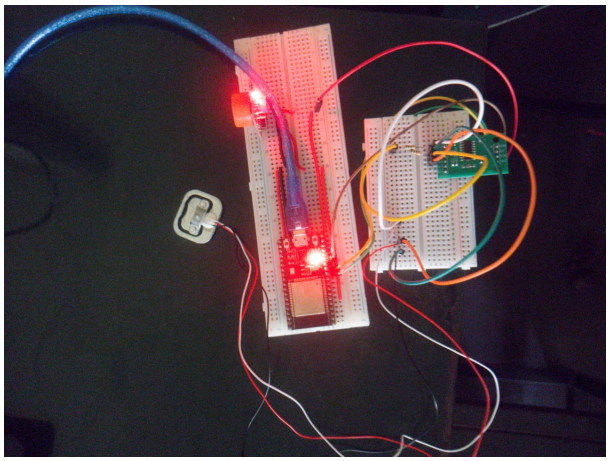
The LPG sensor(MQ6) is an MEMS device that uses a mico-hotplate on a silicon chip together with a metal oxide(SnO_2). The metal oxide has increased resistance as a result of elevated temperature. When methane or other reducing gas agents react with SnO_2, the electrons released into the system by oxidation decrease resistance of the system, providing a signal.

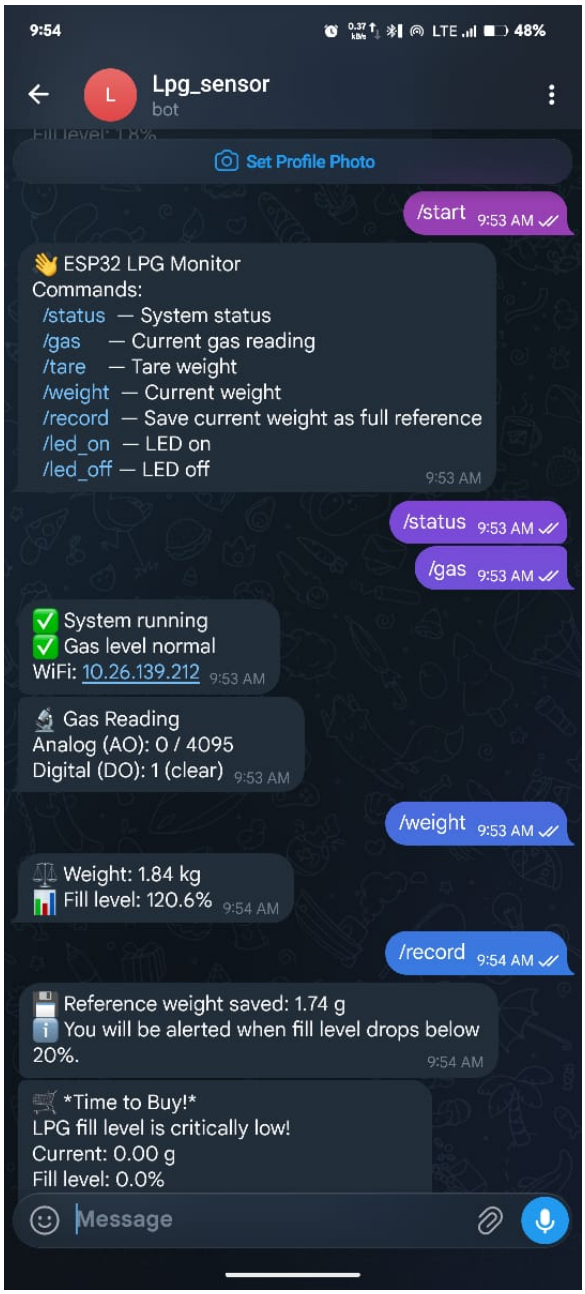
The MQ6 measures the concentration of LPG gas in the air. It outputs an analog signal based on the ppm of LPG present in the air surrounding the sensor. The analog value is converted to digital values with an internal ADC in the ESP32 micro controller. The Esp32 compares the real time value to a threshold value set in the micro controller.

When the real time value exceeds the threshold value, the esp32 notifies the user via telegram to turn off the LPG cylinder. The LPG gas level is monitored for a duration, and return of the base level is also notified to the user.

A Half Bridge 50 kg load cell is used along with Hx711 module to amplify it's signal and the ESP32 measures the value in analog form and converts it to integer range. The weight measurement of the load cell is calibrated with a known weight. After calibration, the weight of the LPG cylinder is saved and a notification is sent to the user when the cylinder reaches 25% of the initial capacity.

Results and Output





Conclusion

This project is an effective and cheap solution to improve LPG usage by real time monitoring of Lpg gas level in the air and real time alerts. The integration of MQ-6 sensor with ESP-32 results in a system to notify lpg gas leak at early stages instantly via Telegram, additionally, the incorporation of a load cell with the HX711 module enables accurate monitoring of cylinder weight, allowing users to track gas consumption and receive alerts when the gas level becomes low.

The system addresses critical safety concerns associated with LPG by reducing response time during leakage events and minimizing the risk of accidents such as fires and explo-

sions. Its low cost, ease of implementation, and remote monitoring capability make it suitable for both domestic and small-scale commercial applications.

Future Scope

The accuracy of the system can be improved by using multiple MQ-6 sensors kept in different location to obtain a more accurate leak. Increasing the no of load cells from 1 to 4 can increase the precision of the weight readings.

Further improvement in the software to incorporate the deep sleep feature of ESP32 can result in long life of the system. Additionally a master and slave configuration settings can be adopted to have these devices used in commercial settings.

Addition of noise filter algorithm can significantly improve the precision of system. A humidity and temperature sensor can be added in addition to decrease false positive alarms. Currently the system relies on the internet to send real-time alerts. For redundancy a sim-card based method can be adopted and offline alter method such as buzzer can be installed.

A tiny machine learning model can be deployed to predict the number of days the gas can be used with the current quantity available and improve visualization can be enabled via a clound service.

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